

Qatar-2b

R-1553

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_200219 · created 2026-08-01T20:13:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458898.86705 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.064
Transit depth (Rp/Rs)^2	3.66 +/- 2.46 [%]
Orbital Inclination (inc)	85.07 +/- 1.72
Airmass coefficient 1 (a1)	1.19 +/- 0.71
Airmass coefficient 2 (a2)	-0.01 +/- 0.26
Scatter in the residuals of the lightcurve fit is	68.41 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	7.5
Transit Duration (day)	0.0683 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.064 vs 0.16327 (deviation 0.43σ)
Duration fit vs published	0.0683 d vs 0.0767 d (deviation 0.87σ)
Mid-time O–C	2.2 min
Depth SNR	3.0
Residual scatter	68.41 %

Light curve

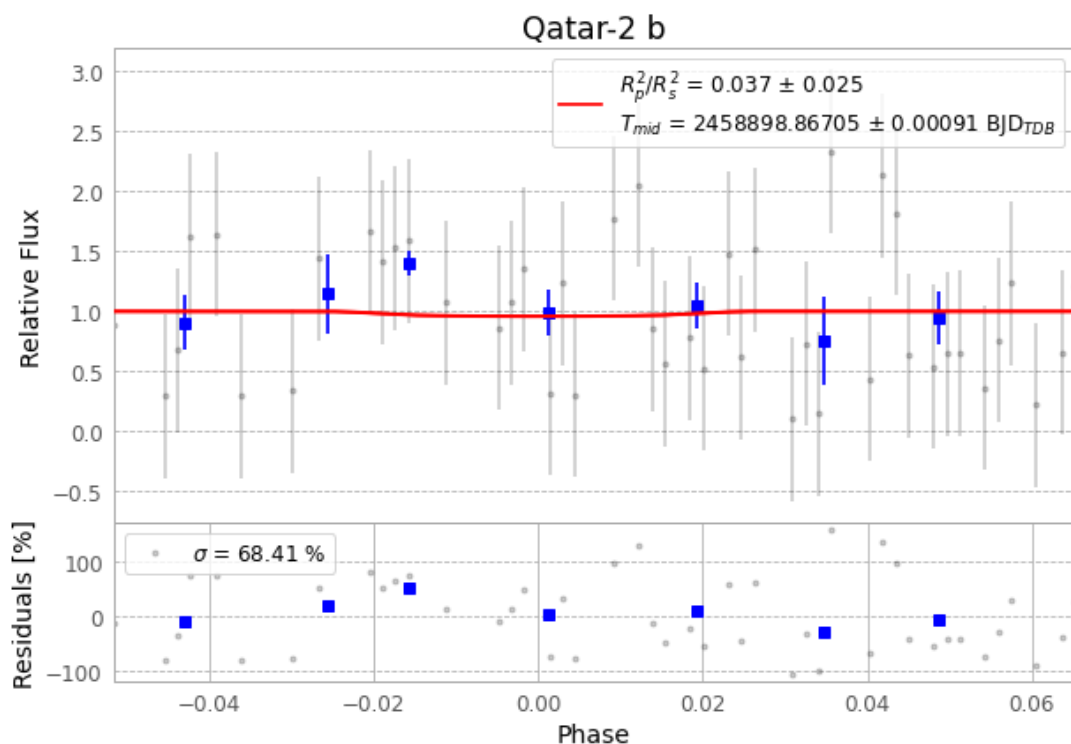


Figure 1 — FinalLightCurve_Qatar-2 b_2020-02-18.png (SHA-256 b8ae22b66b4ce73c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 445], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a1674e910ec71c3a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), Qatar-2200219065712.FITS → Qatar-2200219104812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-200219.log (72cc72d20fea...), FinalLightCurve_Qatar-2 b_2020-02-18.png (b8ae22b66b4c...), FinalLightCurve_Qatar-2 b_2020-02-18.csv (e349069da1f6...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 20:13Z.